# 9.8 Isolating a Quantity of Interest in a Quadratic Equation

## Lesson Introduction

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|---|---|---|---|
| Benchmark | MA.912.AR.1.2 Re-arrange equations or formulas to isolate a quantity of interest. | Learning Targets | • I can rearrange the vertex form of a quadratic equation to isolate a variable of interest.<br>• I can rearrange the factored form of a quadratic equation to isolate a variable of interest.<br>• I can rearrange the standard form of a quadratic equation to isolate a variable of interest. |

| | Building On | Working Toward |
|---|---|---|
| Benchmark Coherence | MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides. | |

| | |
|---|---|
| Essential Questions | • How can I apply the procedures for solving an equation to isolate a variable of interest?<br>• How does the variable of interest selected change the steps in isolating it? |
| Lesson Narrative | In this lesson, students will continue their journey of solving quadratic equations. Students will be solving for a specific variable. They will do this with quadratic equations in vertex form, factored form, and standard form. |

| Component Summary | | | | |
|---|---|---|---|---|
| | 9.8.1 Warm-Up (~5 min.) | Bell Work on solving a quadratic equation by factoring *(SE p. 98)* | 9.8.3 Collaboration (10 min.) | Collaborate on isolating the variable "$a$" in a given quadratic equation *(SE p. 100)* |
| | 9.8.2 Collaboration (15 min.) | Different forms of quadratic equations collaborative activity *(SE p. 99)* | 9.8.4 Guided Instruction (10 min.) | Introduction to isolating the variable "$x$" in a quadratic equation *(SE p. 101)* |
| | | | 9.8.5 Wrap-Up (~5 min.) | Ticket Out the Door on solving a quadratic equation for a specific variable *(SE p. 102)* |

| | Glossary | Materials | Additional Preparation |
|---|---|---|---|
| Resources | Key Terms:<br>• Factored form<br>• Standard form<br>• Vertex form<br><br>Additional Terms: N/A | • N/A | • Work out all the questions ahead of time prior to instruction. Highlight the potential misconceptions that students may have. Isolating variables in a quadratic equation will be difficult for students. Look for ways to add additional scaffolding for your students. |

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td><ul><li>Students may struggle with solving for a variable in isolation.</li><li>Students may apply rules of combining like terms to terms that are not like terms.</li><li>Students may make calculation errors when solving.</li></ul></td><td><ul><li>Students have re-arranged linear formulas for a specified variable. Consider making a connection to what they have learned about linear literal equations.</li></ul></td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns<br>9.8.2 Collaboration provides students with a collaborative opportunity to work with a partner about the various forms of quadratic equations. The discussion questions within this activity provide students the opportunity to discuss their answers, processes, and observations to come to a consensus.</td><td>MA.K12.MTR.4.1: Discussion and Reflection<br>9.8.2 Collaboration provides students a collaborative opportunity to engage in discussions about the various forms of quadratic equations. Students will articulate their mathematical thinking and engage in dialogue about the mathematical thinking of others.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider creating two mixed-partner groups for the 9.8.2 Collaboration and the 9.8.3 Collaboration. Each student would get a different partner for each of the activities so they can get a mixed-perspective experience.<br>9.8.2 Collaboration implements both auditory and visual entry points into discussion. Consider providing roles for students that best fit their learning style as an entry point for the activity or turn your room into a coordinate plane for kinesthetic learners.<br>The 9.8.5 Wrap-Up for this lesson provides a Ticket Out the Door that assesses a student’s level of mastery with solving a quadratic equation for a specified variable.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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# 9.8.1 Warm-Up: Bell Work

***Component Overview:*** *Students will complete a Bell Work activity on solving quadratic equations by factoring. Since this is based on a skill they learned in the last lesson, students should complete this Warm-Up independently.*

## 9.8.1 Warm-Up: Bell Work

1. Solve $2x^2 - 12x + 14 = 2x - 6$ by factoring.

$2x^2 - 14x + 20 = 0$
$2(x^2 - 7x + 10) = 0$
$2(x - 5)(x - 2) = 0$
$x - 5 = 0 \qquad x - 2 = 0$
$x = 5 \qquad x = 2$

*This Warm-Up is based on what students learned in a prior lesson. Since the content is not new, this can provide data to let you know if students retained the information and where they may need additional support.*

# 9.8.2 Collaboration: Forms of Quadratic Equations

***Component Overview:*** *Students will collaborate on the different forms of quadratic equations. Begin this activity by reading the information prior to question 1 out loud with the students. For question 1, students will work with their partner to complete the information in the table. For question 2, read and review the instructions since each partner will work with another classmate to complete the table. For question 3, have students answer independently and then share with the whole class.*

## 9.8.2 Collaboration: Forms of Quadratic Equations

The three different forms of quadratic equations are helpful in providing information about key features of the parabola. It may be helpful at times to understand how a quadratic equation can be rewritten to feature a specific variable.

**Forms of Quadratic Equations**

Vertex form of a quadratic equation is represented as $y = a(x-h)^2 + k$, where $(h, k)$ is the vertex.

Factored form of a quadratic equation is represented as $y = a(x-p)(x-q)$, where $(p, 0)$ and $(q, 0)$ are the $x$-intercepts.

Standard form of a quadratic equation is represented as $y = ax^2 + bx + c$, where $(0, c)$ is the $y$-intercept.

1. Work with your partner to complete the table.
   *Answers in will vary.*

| Equation | Variables of Interest | Variables that Look Easy to Isolate |
|---|---|---|
| $y = a(x-h)^2 + k$ | $a, h, k$ | $a, k$ |
| $y = a(x-p)(x-q)$ | $a, p, q$ | $a$ |
| $y = ax^2 + bx + c$ | $a, b, c$ | $c$ |

*What key features of a quadratic are you confident you could find using an equation? What key features do you find are hard to determine from an equation?*

*Consider creating a visual guide to the forms of quadratic equations. This could include an anchor chart in the classroom or foldables for each student.*

*Challenge students to identify which form of a quadratic equation is presented in each of the equations of the table and provide an explanation.*

# 9.8.2 Collaboration: Forms of Quadratic Equations (continued)

2. Each partner should ask two different classmates which variables they selected. Record their variables and write your summary of their reasons. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

| **Variables That Look Easy to Isolate** | **My Summary of their Reason** | **Initials** |
|---|---|---|
| *Answers will vary. $k$.* | *Answers will vary. $k$ is the easiest because it's by itself.* | |
| *Answers will vary. $a$.* | *Answers will vary. $a$ is the easiest because once the binomials are divided it's left alone.* | |

3. The variable $a$ is the only variable that is common to all three forms. Which form appears to be the easiest to use to isolate the variable $a$?
*Answers will vary. Factored form, $a(x-p)(x-q)$, appears to be the easiest because to get 'a' isolated, the only operation needed is division.*

*Encourage students to share their rationale with you as you circulate. Allow students to use explanations such as "it looks easy," but probe them for why another variable does not look easy.*

# 9.8.3 Collaboration: Isolating the Variable $a$

***Component Overview:*** *Students will collaborate to isolate the variable "$a$" in a given quadratic equation.*

## 9.8.3 Collaboration: Isolating the Variable $a$

1. Work with your partner to complete the steps for isolating the variable $a$.

| Vertex Form | $y = a(x-h)^2 + k$ |
|---|---|
| Subtract $k$ from both sides. | $y - k = a(x-h)^2 + k - k$<br>$y - k = a(x-h)^2$ |
| Divide both sides by $(x-h)^2$. | $\frac{y-k}{(x-h)^2} = \frac{a(x-h)^2}{(x-h)^2}$<br>$a = \frac{y-k}{(x-h)^2}$ |

| Factored Form | $y = a(x-p)(x-q)$ |
|---|---|
| Divide both sides by $(x-p)(x-q)$. | $\frac{y}{(x-p)(x-q)} = \frac{a(x-p)(x-q)}{(x-p)(x-q)}$<br>$a = \frac{y}{(x-p)(x-q)}$ |

| Standard Form | $y = ax^2 + bx + c$ |
|---|---|
| Subtract $bx + c$ from both sides. | $y - (bx + c) = ax^2 + bx + c - (bx + c)$<br>$y - bx - c = ax^2$ |
| Divide both sides by $x^2$. | $\frac{y - bx - c}{x^2} = \frac{a}{x^2}$<br>$a = \frac{y - bx - c}{x^2}$ |

2. Which form, if any, was the easiest to use to isolate the variable $a$?
*Answers will vary. Factored Form. Standard Form.*

*Have students share their answer to question 3 of 9.8.2 Collaboration as the class hook for this Collaboration.*

*Students may struggle with the actual completion of the steps without numeric values. Consider giving each form to different student pairs and setting a timer for each pair to solve that form for $a$. Then, form groups to verify the work.*

*What made your choice the easier one to use to isolate the variable "$a$"? Which form was the most challenging to isolate the variable "$a$" in?*

# 9.8.4 Guided Instruction: Isolating the Variable $x$

***Component Overview:*** *Students will be provided with Guided Instruction on how to isolate the variable x from vertex and standard form of the equation.*

## 9.8.4 Guided Instruction: Isolating the Variable $x$

1. Complete the table for isolating the variable $x$ in vertex form.

| **Vertex Form** | $y = a(x-h)^2 + k$ |
|---|---|
| Subtract $k$ from both sides. | $y - k = a(x-h)^2$ |
| Divide both sides by $a$. | $\frac{y-k}{a} = (x-h)^2$ |
| Take the square root of both sides.<br><br>When taking the square root of both sides include $\pm$ to represent both of the possible outputs. | $\sqrt{\frac{y-k}{a}} = \sqrt{(x-h)^2}$<br><br>$\pm\sqrt{\frac{y-k}{a}} = x - h$ |
| Add $h$ from to sides. | $h \pm \sqrt{\frac{y-k}{a}} = x$ |

The $\pm$ represents both possible results of a square root. Remember, when asked to find $\sqrt{36}$ it is understood that the principal square root, $6$, is the answer, but that both $6$ and $-6$ are correct because both values can be multiplied by itself to get $36$. This can be written as $\sqrt{36} = \pm 6$.

2. Rewrite $x = -w \pm \sqrt{9y}$ as two equations.

$x = -w + \sqrt{9y}$
$x = -w - \sqrt{9y}$

3. Rewrite $x = h \pm \sqrt{\frac{y-k}{a}}$ as two equations.

$$x = h + \sqrt{\frac{y-k}{a}}$$

$$x = h - \sqrt{\frac{y-k}{a}}$$

*Consider using a gradual release for each step of question 1. For example, explain the first step, explain why that step exists, and show them how to do it. For the second step, explain the step, explain why it exists, and then have students do it. Continue to release to students and go through the whole table.*

*Why do you think that the question is asking for there to be two written equations?*

# 9.8.4 Guided Instruction: Isolating the Variable $x$ (continued)

4. Rewrite $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ as two equations.

$x = \frac{-b+\sqrt{b^2-4ac}}{2a}$

$x = \frac{-b-\sqrt{b^2-4ac}}{2a}$

5. Complete the table for isolating the variable $x$ in standard form.

| **Standard Form** | $y = ax^2 + bx + c$ |
|---|---|
| From standard form the vertex of the parabola can be found using $\left(\frac{-b}{2a}, \frac{c}{a} - \frac{b^2}{4a^2}\right)$. Rewrite standard form to vertex form using this vertex. | $y = a(x-h)^2 + k$<br>$y = \left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2}$ |
| Subtract $\frac{c}{a}$ from both sides. | $y - \frac{c}{a} = \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}$ |
| Add the opposite of $-\frac{b^2}{4a^2}$ to both sides. | $y - \frac{c}{a} + \frac{b^2}{4a^2} = \left(x + \frac{b}{2a}\right)^2$ |
| Take the square root of both sides.<br><br>When taking the square root of both sides, include $\pm$ to represent both of the possible outputs. | $\sqrt{y - \frac{c}{a} + \frac{b^2}{4a^2}} = \sqrt{\left(x + \frac{b}{2a}\right)^2}$<br>$\pm\sqrt{y - \frac{c}{a} + \frac{b^2}{4a^2}} = x + \frac{b}{2a}$ |
| Subtract $\frac{b}{2a}$ from both sides. | $-\frac{b}{2a} \pm \sqrt{y - \frac{c}{a} + \frac{b^2}{4a^2}} = x$ |

*Introducing students to re-writing equations with $\pm$ as two separate equations establishes that the $\pm$ will be two different results. If students learn the quadratic formula as a single formula, the $\pm$ is often misunderstood or overlooked. Questions 2 and 3 are scaffolds to question 4. Then, students are asked to complete steps to gain an understanding of the quadratic formula using vertex form. For many students the algebra will be difficult, but that is not a reason to avoid doing the question. Use methods such as the cover-up method to help students focus on the described step. Refer to the description of the step to focus students.*

*Honors Algebra 1 students could be asked to complete the steps with their partners and to then verify them.*

*Student responses to instruction in this section can provide valuable differentiation data for students who need additional support.*

# 9.8.5 Wrap-Up: Ticket Out the Door

***Component Overview:*** *Students will complete a Ticket Out the Door that assesses their level of mastery of solving a quadratic equation for a specific variable. Have students complete this task independently so you can determine each student's knowledge of the lesson content.*

### 9.8.5 Wrap-Up: Ticket Out the Door

1. Solve $y = 2(x-5)^2 + k$ for $k$.
   $y - 2(x-5)^2 = k$

2. Solve $y = -x^2 + bx + 7$ for $b$.
   $y + x^2 - 7 = bx$

   $$\frac{y + x^2 - 7}{x} = \frac{bx}{x}$$

   $$\frac{y + x^2 - 7}{x} = b$$

*Student performance on these questions can provide data for additional support for students that may be needed. Each question assesses a different component of the content of the lesson.*

*Question 1: Isolating in vertex form*

*Question 2: Isolating in standard form*